# Agile Sliding-Spotlight SAR Satellite Scheduling with Per-Task Resolution Requirements: The Coverage Cost of Tightening the Order

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**Abstract**

Agile Earth-observation satellites that steer a single antenna across stripmap, sliding-spotlight, and spotlight modes are usually scheduled under an exogenous, fixed imaging duration. For a sliding-spotlight platform that assumption fails in a quantifiable way: the dwell is produced by the mode continuum and by the selected start, and it enters the inter-task transfer constraint that decides what a pass can carry. This paper makes the dwell endogenous and carries the order resolution requirement as a per-task hard constraint: the mode parameters follow the order where the mode has them, so that the resolution delivered at the selected start equals the order, while the stripmap tier stays at its antenna floor. Served priority is the only objective; a fulfillment ratio, computed on the library quality caliber and averaged over all tie-set members, is a reported reading and not an optimization target. A frozen nine-point demand grid scales every order from 0.8 to 2.0 without regenerating the scenario population or re-routing modes, and three shape criteria (monotone, saturation, tight arm) are frozen in advance and reported with their failures. The dense 100-task family R3H pays a coverage cost at the order point of 4.15 percentage points against its own saturation. The single-orbit family R1C meets a floor step: at $s = 0.8$ it admits no feasible individual in 10 of 10 seeds. R1C at $s = 0.9$ reads a median

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coverage of 0.5398. The 50-task family R2F pays a coverage cost at the order point of 2.36 percentage points against its own saturation. Over the nine points 477,308 task instances are served, with no delivery-identity violation recorded by the evaluator. Conclusions are scoped to order sensitivity under a fixed mode assignment.

*Keywords:* agile Earth-observation satellite scheduling, synthetic aperture radar, sliding spotlight, per-task resolution requirement, coverage cost of a tighter order, demand sensitivity

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## 1. Introduction

### *1.1. Motivation*

Agile Earth-observation satellites steered by a single antenna can operate their synthetic aperture radar (SAR) in a continuum of imaging modes: stripmap, sliding spotlight, and spotlight are one physical mode family parameterised by where the beam's virtual rotation centre sits. Which point of that continuum a task receives is not a free stylistic choice — it is fixed by the requested resolution, the scene, and the geometry of the pass — and it determines how long the satellite must dwell on the scene. Scheduling therefore cannot treat the imaging duration as an exogenous constant, as the classical agile-scheduling literature largely does [24, 12, 13]: in sliding-spotlight operation the dwell is produced by the imaging physics, it changes from task to task, and it feeds back into how many tasks a pass can carry and at what image quality. In that literature the resolution requirement, when it appears at all, is an attribute of the request that is settled before the schedule is built: the observation duration is a task attribute fixed in advance [19, 12], and on the SAR side the required resolution already fixes the synthetic-aperture time and steering span of a designed observation [11].

This paper studies that coupling on a single satellite in a regime of single-pass dense task clusters under true orbit geometry. Scheduling decisions select which tasks to serve and when to start imaging within each task's visibility windows;

the mode and its parameters follow from the product order, and the dwell, the maneuver, and the image quality follow from the mode and the window. The systematic study of scenario density as an experimental axis is outside the scope of this paper; the density ladder is fixed at registration (Section 5.2).

Two discipline risks shape the methodology as much as the physics does. First, a measured effect on coverage can be a property of the *search* rather than of the *scenario*: in this world the greedy anchor already attains the family maximum in 258 of the 260 sweep units, so the coverage response reported here is produced by the mode and parameter adaptation rather than by the optimiser's ability to find it. Second, quality statements are only as good as their measurement calibers: several formally plausible formulations (an ambiguity sum that includes the principal image, a range resolution that varies per task, a squint-dependent routing resolution) are registered as not used, and the paper says so where it matters.

### *1.2. Contributions*

Following the registered narrative frame — registered here meaning fixed in the project's preregistration before drafting began — the paper's contributions are three: the platform that carries the order into the physics, the measured coverage cost of tightening that order, and the discipline that keeps both claims testable:

1. **An enabling physics-to-scheduler platform** (the platform leg). A three-mode continuum formalisation (Section 3) that carries the mode choice, the scene-level dwell closures, the order resolution as a per-task hard constraint, and the maneuver-coupled feasibility predicates into one scheduling loop: the imaging mode is resolved from the product order, the mode parameters follow the order where the mode has them so that the delivered resolution equals the order at the selected start, and the dwell duration is a consequence of the order rather than a constant.
2. **The coverage cost of tightening the order** (main result). Three scenario families swept over a frozen nine-point demand grid at a single

search budget and under a frozen mode assignment: a floor step on the stripmap family, a non-monotone response on the 50-task family whose two dips are traced to the greedy anchor construction, order-point costs of 4.15 and 2.36 percentage points against each family's own saturation, and 477,308 served task instances with no delivery-identity violation (Section 6).

3. **A discipline that keeps the result falsifiable** (methodology). Shape criteria frozen before this scenario world's data were read and reported with their failures rather than re-tuned to the data; caliber labels that keep the two quality calibers apart and keep the fulfillment ratio a reading; and an explicit separation of the search face from the construction face — the anchor is coverage-optimal in 258 of 260 units, so the coverage response is a construction result, not a search finding (Sections 5 and 6).

Throughout, the trade-off the study claims is the resolution-versus-dwell-time axis of the mode continuum; the antenna-design question of resolution against swath is a sensor-layer consequence that this simulation does not model, and it is treated as a limitation with future work (Section 7.1). The serial-bound reading of how many dwell-coupled tasks a pass carries is stated with a per-pass caliber and without a number, because its serial bound has not been recomputed in this scenario world; it is carried as a scope statement (Section 7.1) rather than as a result.

The move this paper makes is to take the resolution off the request and put it into the constraint set: the order is what the scheduler must satisfy task by task, the mode parameters are what carry it into the physics, and the coverage loss is what the paper measures. The claim is scoped to the literature surveyed below: the survey rests on the project's curated literature pool — a converted and catalogued bibliography of the field plus an incremental paper-search sweep over the months preceding writing, whose nearest neighbours Section 2 engages directly. What the tested coupling in fact produced in this scenario world is reported in Section 6: the coverage–demand curves of the three families, the

floor step and the two anchor-carried dips, the order-point costs against each family's own saturation, and the demand-satisfaction count — the platform's value is that each of these readings is registered and reproducible.

### *1.3. Paper organisation*

Section 2 positions the work along sliding-spotlight imaging, agile-satellite scheduling, and quality-aware scheduling. Section 3 formalises geometry, modes, dwell, the order as a per-task constraint, and the demand-tightening axis. Section 4 describes the solver. Section 5 presents the scenario world and the frozen analysis design; Section 6 reports the coverage–demand curves and the demand-satisfaction readings; Section 7 discusses mechanisms, literature contrasts, and the full limitation set; Section 8 concludes.

## 2. Related Work

This section positions the paper along three lines: sliding-spotlight SAR imaging (where the dwell physics lives), agile Earth-observation satellite scheduling (where the fixed-dwell assumption lives), and quality-aware scheduling (where image quality enters planning decisions).

### *2.1. Sliding-spotlight and spotlight SAR imaging*

The sliding-spotlight family is standardly formulated as steering about a fixed virtual rotation centre: the TerraSAR-X extended chirp scaling processor is built on that geometry [14], the baseband azimuth-scaling processor of Prats et al. handles both sliding spotlight and terrain observation by progressive scans (TOPS)[21]; in sliding spotlight the rotation centre lies beyond the swath, in TOPS it lies behind the sensor. Kim et al. derive the attitude-maneuver and saturation-avoidance time formulas for squint staring and sliding spotlight on the same geometry [16]. Operational descriptions sometimes state the steering as a constant angular rate — Zhang et al.'s Gaofen-3 (GF-3) account does so explicitly and describes it as a small-angle simplification of the same geometry [27] — and processing work that pushes to high squint is likewise explicit that

the linearised steering assumption breaks there [3, 20]. This paper adopts the rotation-centre formulation throughout (Section 3.1); its scheduling windows reach squints where only the exact geometry is valid, which is why the two readings must not be mixed.

Two quantitative anchors from the imaging line matter to this study, and both are quoted as their own platforms' literature values. First, Kim et al.'s Table 2 defines a $\pm 12^\circ$ task envelope for its reference mission; the design envelopes registered in this paper — $\pm 45^\circ$ for stripmap/spotlight and $\pm 26.9^\circ$ (a registered conservative value, a design choice) for sliding spotlight — are separate fields and are never merged with the literature task domain [16]. Second, Song et al. report for their sparse-imaging platform that the azimuth ambiguity-to-signal ratio is anti-correlated with the pulse repetition frequency (PRF) and degrades as the scan angle grows (by about 8.6 dB beyond $1^\circ$ of scan for conventional sliding spotlight, and about 15.2 dB for the sparse variant), alongside a range-ambiguity reduction of about 10.7 dB for the sparse mode [22]; as Section 7.2 shows, the direction of the ambiguity-versus-squint dependence in our model is the opposite of that literature anchor, and the contrast is a result of this paper rather than a contradiction to be smoothed over. The standard ambiguity and resolution relations otherwise follow the textbook treatment [2].

The dwell time itself is treated in the imaging literature as a deterministic function of mode and geometry — Kim et al.'s saturation-avoidance time chain is exactly that [16] — but it stops at the imaging problem boundary: in the imaging references this paper draws on, the scheduling layer above it is not part of their treatment.

### *2.2. Agile Earth-observation satellite scheduling*

The agile-scheduling literature is broad and mature: hybrid genetic-simulated-annealing planners for moving-target observation [12], deep-reinforcement-learning planners for large mission-planning instances [24, 19, 6], adaptive large-neighborhood search with learned guidance or with neighbourhood repair [25, 17], constraint-programming formulations for super-agile platforms [18], memetic and multi-

chromosome cooperative planners [1, 23], and profit models in which the observation value depends on the interval between revisits [13, 7]. Across the formulations surveyed here, one structural assumption is near-universal: the imaging duration of a task is an exogenous constant of the task, known before the scheduler runs. The scheduler then packs fixed-length intervals into time lines, and whatever the objective — profit, coverage, latency — the duration never feeds back.

For an agile SAR platform in sliding-spotlight operation that assumption is false in a specific, quantifiable way: the dwell is produced by the mode physics (the tan-exact closure chain of Section 3.2), it varies per task with the assigned mode and the squint of the selected start, and it enters the inter-task transfer constraint that decides what else the pass can carry. This paper makes the dwell endogenous, carries the order resolution as a per-task constraint on that dwell, and measures what tightening the order costs in coverage.

### *2.3. Quality-aware scheduling*

Image quality reaches the planning literature in three roles, and the difference between them and this paper is worth stating plainly. Image-quality research on the SAR side carries a substantial assessment literature — fidelity metrics, quality checkers, and radiometric corrections applied to formed images [2, 14, 21]. Imaging-mode design starts from a resolution requirement and trades it against swath [22], and mission-level SAR planning computes the required synthetic-aperture time and steering span from the required resolution and the incidence angle [11]. Multi-objective Earth-observation and SAR mission studies optimise more than one criterion explicitly alongside coverage or priority [26, 9, 11]; that line of work exists, it is not the line this paper takes, and nothing here claims its absence. What changes here is the *placement* of quality. It leaves the objective set: the per-mode index of Section 3.3 states a platform capability and supplies the numerator of a reported fulfillment ratio, and it carries no selection pressure on the search.

### *2.4. Positioning*

The three lines meet at a seam rather than at a gap, and the seam is narrow. The imaging line supplies the resolution-to-aperture chain and stops at the imaging problem boundary: a required resolution and incidence angle fix the synthetic-aperture time and steering span of a designed observation [11], and the sliding-spotlight processors deliver such a design at a fixed rotation centre [14, 21]. The scheduling line supplies the packing machinery for tasks whose duration is an attribute fixed before the schedule is built [19, 12]. The quality-aware line supplies assessment calibers and mission-level weightings [2, 9]. This paper puts the order resolution into the scheduling constraint set: the mode parameters follow the order where the mode has them, the resolution delivered at the selected start equals the order, and the coverage cost of tightening the order is measured on a frozen grid under a fixed mode assignment (Section 3 and Section 6). Sections 3–6 build that problem on a registered, preregistered footing — the demand grid, the shape criteria and the caliber labels are fixed before this world's data were read — and report what the tightening costs.

## 3. Problem Formulation and Scheduling Model

This section formalises the observation geometry, the three-mode imaging library, the dwell-duration closures, the per-mode image-quality model, and the constraints and objectives of the scheduling problem.

### *3.1. Observation geometry and imaging modes*

We adopt the X-band reference platform of Kim et al. [16]: altitude $h = 570$ km, inclination $45^\circ$, wavelength $\lambda = 0.031$ m, azimuth beam width $\theta_a = 0.36^\circ$, and a square scene of 5 km side length. The derived geometry is $D = \lambda/\theta_a = 4.9338$ m (antenna azimuth length), aperture-centre slant range $R_0 = 775.4$ km, and inertial scene velocity $v_s = 7578.05$ m/s. This is the platform inertial velocity, registered as the scene-traversal velocity of this scenario world rather than the beam-footprint ground velocity; it enters every dwell closure

and the Doppler bandwidth as the single frozen value, so cross-mode and cross-squint dwell ratios are invariant to the choice, while absolute dwell durations and every reading that mixes dwells with velocity-independent constants (the coverage readings) or evaluates the ambiguity ratio at a fixed PRF carries the convention. Each task is a scene; the scheduled quantity is *when* to image it, not *whether* to reconfigure the payload.

The three imaging modes — stripmap (STR), sliding spotlight (SLS), and spotlight (SPT) — are treated as a single-parameter continuum over the position of a fixed virtual rotation centre [16, 27, 22]: writing $k$ for the rotation-centre parameter and $A = 1 - k$ for the sliding factor, $k \to 0$ places the rotation centre at infinity (stripmap, parallel antenna axis), $0 < k < 1$ places it below the surface beyond the scene (sliding spotlight), and $k \to 1$ places it at the target (spotlight). The registered operating point of this study is $k = 0.797317$, i.e. a rotation-centre-to-scene-centre distance $r_0 = 197\,119$ m $\approx 197$ km. We stress the symbol discipline that the two distances are dimensionally identical but semantically distinct: $r_0 \approx 197$ km is the rotation-centre-to-scene-centre distance, whereas $R_0 \approx 775.4$ km is the aperture-centre slant range; their ratio is $r_0/R_0 = 0.254207$, and the two are never interchangeable (Fig. 1).

The sliding factor: under the rotation-centre semantics adopted here, $A = r_0/(R_0 + r_0)$ [27]; the form $A = 1 - R_0 k_\theta / v_g$, with $k_\theta$ the azimuth steering rate and $v_g$ the ground speed (Song Eq. (1)) [22], is a restatement within the same rotation-centre model — for the steering rate that places the rotation centre at $r_0$ — not an independent literature identity. The per-mode azimuth resolutions at zero squint are $\rho_{a,\mathrm{STR}} = D/2 = 2.4669$ m, $\rho_{a,\mathrm{SLS}} = A\,D/2 = 0.5000$ m, and $\rho_{a,\mathrm{SPT}} = \lambda/(2\beta_{\mathrm{span}}) = 0.3000$ m with $\beta_{\mathrm{span}} = 2.9603^\circ$ the spotlight sweep span (library design constant).

Under the rotation-centre semantics the spectral compression factor is a constant at every squint. Presenting the Doppler rates in the aperture-range form,

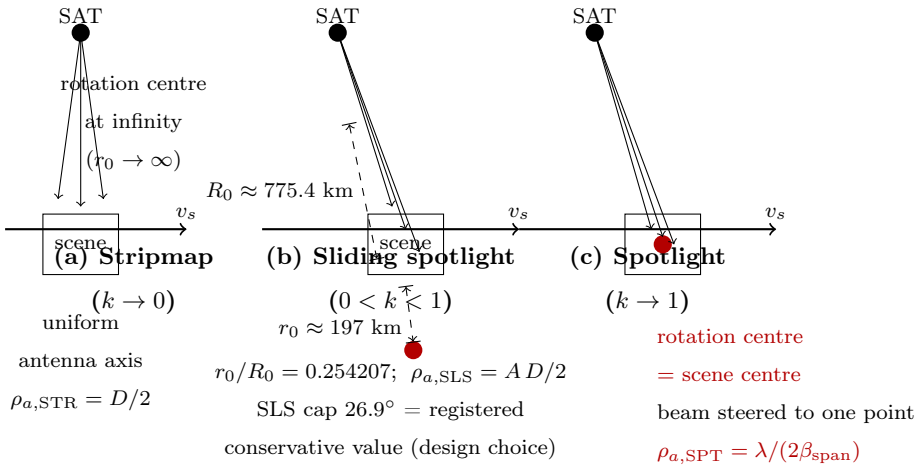
$$K_a = \frac{2v_s^2 \cos^3 \bar{\psi}}{\lambda R_0}, \qquad K_c = \frac{2v_s^2 \cos^3 \bar{\psi}}{\lambda\,(R_0 + r_0)}, \qquad a_{\mathrm{res}} = 1 - \frac{K_c}{K_a} \equiv A,$$

the $\cos^3$ factors cancel at beam-centre crossing, so $a_{\mathrm{res}}$ does not degrade with squint; the sliding-spotlight design azimuth resolution is the constant $A\,D_a/2$ [27, 22].

The registered squint domains are separate registered fields, listed apart from the literature task envelope of Kim et al. ($\pm 12°$, their Table 2 [16]): STR and SPT carry a design domain of $\pm 45°$, and SLS carries a registered conservative envelope of $\pm 26.9°$. The SLS cap is a *registered conservative value (design choice)* — antenna steering margin plus frozen-data continuity — and not a physical boundary, carried as a design choice under those two considerations with no analytic derivation claimed: under the rotation-centre semantics $a_{\mathrm{res}} \equiv A > 0$ at every squint, so high-squint reachability is limited by the registered cap rather than by mode physics. Two extrapolation disclosures attach to the $45°$ design domain: the azimuth-resolution projection correction (resolution-ellipse rotation) is outside the closed form, and the $45°$ dwell readings of STR/SPT are extrapolated, because the planar tan-difference closure omits Earth curvature against the full-curvature $R_0$ (Section 7.3).

### *3.2. Dwell duration and visibility windows*

The dwell of a task is the saturation-avoidance time (SAT) at scene level: the time the beam needs to sweep the full square scene at the task's squint, where $W_a$ denotes the scene (square) size — 5 km on the reference platform — entering as a scene-length traversal term. The production closures are the tan-exact Kim Eq. (22)/(24) chain [16]. For STR the scene dwell is the exact tan difference of the beam-edge crossing angles plus the scene traversal $W_a/v_s$; for SLS it is the symmetric construction of Eq. (22)/(24) with the sliding factor in the denominator, evaluated at the SLS design azimuth resolution rather than the requested task resolution (conservative for coarser demands); for SPT it is the exact Eq. (10)–(11) tan-difference with the library sweep span. A planar closed form with the $\sec^2 \bar{\psi}$ squint correction, $T_{\mathrm{SLS}} = (R_0\theta_a + W_a)/(A\,v_s)\cdot\sec^2\bar{\psi}$ (scene size inside the correction), survives as the test-time oracle: at the Kim reference squint of $12°$ it deviates from the production chain by $+0.4185\%$, and




Design squint envelopes (separate registered fields): STR/SPT ±45° (design domain; 45° dwell readings are extrapolated); SLS ±26.9° (registered conservative value, design choice); Kim ±12° = literature task envelope [16]. Under the rotation-centre semantics $a_{\mathrm{res}} \equiv A > 0$ at every squint: high-squint reachability is limited by the registered envelope, not by mode physics.

Figure 1: Three modes as a single-parameter continuum over the virtual rotation centre (azimuth squint plane, schematic). (a) Stripmap: rotation centre at infinity, uniform antenna axis. (b) Sliding spotlight: rotation centre beyond the scene ($k = 0.7973$); $r_0 \approx 197$ km = rotation-centre-to-scene-centre distance, $R_0 \approx 775.4$ km = aperture-centre slant range, $r_0/R_0 = 0.254207$; the SLS cap 26.9° is a registered conservative value (design choice), not a physical boundary. (c) Spotlight: rotation centre at the scene centre. Design squint domains listed as separate registered fields: STR/SPT ±45° (design domain; 45° dwell readings are extrapolated), SLS ±26.9° (registered conservative value, design choice), and the ±12° literature task domain of Kim et al. [16].

the deviation grows to +2.1% at the SLS cap and +5.3% at 45° (an extrapolated-domain reading, Section 3.1), which is what triggered the kill-test switch from the planar form to the tan-exact chain in the first place. A frozen window-tail pad of 1% of the dwell is carried as a conservative margin on the production path; it is not claimed to bound the production closure's own squint error, which is not separately quantified here. Two provenance notes complete the closure.

First, the Kim Eq. (22)/(24) chain is a zero-Doppler-clamped (broadside) attitude construction; applying it at nonzero squint is a registered approximation, and the growth of the planar oracle's deviation with squint, quantified above, bounds the oracle's linearisation error rather than the production chain's — it is evidence that the broadside-derived closure carries a modelling gap rather than numerical noise. Second, the resulting dwell feeds the inter-task transfer requirement (constraint C2, defined in Section 3.4) that determines pass capacity, which is why the deviation pattern and the frozen window-tail pad are carried with the constraint semantics (Section 7.3) rather than with the numerics.

The zero-squint anchors close an identity: $\theta_a/A = \lambda/(2\rho_a)$ holds at the library design point ($0.006283185307/0.202683397006 = 0.031000000000$ rad $= \lambda/(2 \times 0.5\,\mathrm{m})$) with a full-precision residual of $3.5 \times 10^{-18}$. The surviving guard of this identity is a dedicated regression test in the duration-model suite.

Each task carries one or more visibility windows produced by ground-track pass enumeration over the scenario span; a window fixes the closest-approach epoch $t_{\mathrm{ca}}$, the per-window slant range, and the earliest/latest feasible imaging start. The decision variable $\tau \in [0, 1]$ of a selected task is mapped onto the feasible start interval of its window (Section 4). The squint at the selected start, $\bar{\psi}$, is measured at the window's geometry.

One scheduling-level geometric quantity needs a definition here because it enters the inter-task constraint: the maneuver angle between consecutive tasks $i, j$, $d_\psi = |\psi_i - \psi_j|$, is the difference of their start squints. This is a planar-caliber *lower bound* on the true attitude excursion: it omits the sweep *during imaging* (the antenna keeps steering at 0.4233–0.6686 °/s, i.e. 1.7–2.7° over a 4 s dwell) and the start-squint difference understates the off-vertical attitude compression by 15–25%. The full disclosure of the two-sided understatement and the margin-pad argument is given in Section 7.3.

*3.3. Image quality model*

The per-task quality index is an inverse-pixel-area form with a range-normalised (non-normalised, $m^{-2}$ caliber) readout,

$$f_{q,i} = \frac{1}{\rho_{\mathrm{rg}}\,\rho_a(m_i)}\; g_T(m_i)\left(\frac{R_0}{R}\right)^3,$$

where $\rho_{\mathrm{rg}}$ is the ground-range resolution, $\rho_a(m_i)$ the zero-squint azimuth resolution of the assigned mode, and $g_T$ the transmit-path antenna gain term. The $(R_0/R)^3$ factor carries the range dependence of the radiometric term: two-way spreading falls as $R^{-4}$ while the resolution cell grows linearly with range, so the noise-equivalent backscatter referenced to that cell scales as $R^3$, and the factor is therefore a *relative* normalisation carried without calibration; at $R = R_0$ the identity chain collapses to $f_q = (g_T/\rho_a)\cdot(1/\rho_{\mathrm{rg}})$ [22]. A scope note on the name: the index is dominated by its azimuth-resolution term, and no *calibrated* radiometric quantity — noise-equivalent sigma zero, signal-to-noise ratio, or radiometric resolution — enters the index: the only radiometric-shaped factor is the uncalibrated relative range dependence above, and it is evaluated at each window's own slant range, so the index of a task depends on which of its windows is scheduled (Section 7.3); the ambiguity ratio enters separately as an admission constraint, not as a traded objective. "Image quality" in this paper therefore denotes spatial-resolution quality under a gain-shaped multiplier.

The index has two roles in this paper, and neither of them is an optimization target. It states a platform capability, and it supplies the numerator of a reported reading. As a capability statement, at the zero-squint library caliber the per-mode constants are 0.8107 (STR), 19.7352 (SLS) and 54.8322 (SPT) $\mathrm{m}^{-2}$: magnitude ratios of 24.3× (SLS over STR), 67.6× (SPT over STR) and 2.8× (SPT over SLS). Because the squint factor cancels inside the gain term (exactly for STR and SLS, and to within 0.07% for SPT across the full envelope), these constants — not squint-dependent readings — are what separates the modes on this axis, and they are properties of the mode library rather than results of a schedule.

The reported reading is a fulfillment ratio. For a served task it is the $f_q$ at the realised window and the selected start's squint, divided by the best $f_q$ the task could reach over all of its windows at zero squint, so the ratio is at most one. It is computed on the library caliber, averaged over all tie-set members rather than read off the anchor solution, and reported as a reading with no optimization role. Its label travels with every number it states: *[library caliber $f_q$ | per-task $f_q$ at the realised window and squint over the zero-squint optimum over all windows | mean over all tie-set members | reported reading, not an optimization target].*

A descriptive delivered variant is registered beside it (formula-spec §8.17-10):

$$f_{Q,\mathrm{deliv}} = \frac{D}{2\,\rho_{\mathrm{req}}^2 \cos^2\psi_{\mathrm{sel}}\,\rho_{\mathrm{rg}}}.$$

It is a descriptive variant, it is not comparable with the library caliber, and it is never placed in the same table as that caliber. It also has no stripmap domain: stripmap delivers the library floor and a task-level mode-parameter request on that mode raises (`duration.py:216-219`).

No directional claim is registered between coverage and the fulfillment ratio. Only the coefficient and its sample size are reported, with the caliber named on the same line; the sign is an observation of this world's data, not a property of the mode continuum.

Three caliber statements are frozen with this definition. First, $\rho_{\mathrm{rg}}$ is held at the design-anchor constant of 0.5 m (registered at 45° incidence, scene-centre semantics); it is not recomputed per task, and the magnitude narrative carries a limitation (Section 7.3). Second, $g_T$ is *self-derived* and *internally verified*: no external literature source carries the noise-equivalent sigma zero (NESZ) equation set for this mode family, so the verification route is the internal identity chain above (and its regression guard), and the absence of external cross-validation is disclosed. Third, the squint-dependent achieved resolution $\rho_{a,\mathrm{ach}}(m,\bar{\psi}) = \rho_a(m,0)/\cos\bar{\psi}$ is registered for reporting; the router consumes the zero-squint constant, and the only bandwidth consumer of $\rho_{a,\mathrm{ach}}$ is the Doppler bandwidth $B_D = v_s\cos\bar{\psi}/\rho_a(m,0)$ — there is no consumer of the

achieved resolution in mode routing. The projection relation behind the anchor is $\rho_{\mathrm{rg}} = c/(2\,B_{\mathrm{rg}}\sin\eta_i)$ with the range bandwidth $B_{\mathrm{rg}}$ frozen at the design incidence; the planar evaluation layer holds the incidence at that design value, which is why $\rho_{\mathrm{rg}}$ enters the objective as a constant, with the incidence-band variation disclosed separately in Section 7.3.

The azimuth ambiguity-to-signal ratio (AASR) is defined with the ambiguity sum *excluding* the *zeroth* (principal-image) term,

$$\mathrm{AASR} = \frac{\sum_{m\neq 0}\int_{-B_D/2+f_{ac}}^{B_D/2+f_{ac}} \left|G\big(f_a + m\cdot\mathrm{PRF}\big)\right|^2 df_a}{\int_{-B_D/2+f_{ac}}^{B_D/2+f_{ac}} \left|G(f_a)\right|^2 df_a},$$

following the standard convention [2]: keeping $m = 0$ would place the principal image in its own numerator and force AASR $\geq 1$ identically. The frequency-axis rescaling by $(1 - K_c/K_a) \approx (1 - k_\theta R_0/v_g)$, a flat-ground bridge between the $K_c$ and $v_g$ forms per the frozen formula registration, under rotation-centre steering is what couples the steering geometry into the ambiguity integral [22]. The displayed integral is Song's general form, with the absolute-centre shift $f_{ac}$ carried in the integration limits; the implementation integrates the same sums about the DC-relative coordinate, where that shift cancels and the absolute centre drops out (Section 7.3). A footnote on the spotlight special case: Song et al.'s spotlight AASR [22] is the degenerate integral of the same form in which neither numerator nor denominator contains $f_a$ after the rescaling, which is why the SPT ambiguity in our model is squint-invariant. Integration caliber: the ambiguity sum runs over the azimuth-frequency axis alone — the two-way pattern enters as a function of azimuth frequency, with no cross-range (elevation-plane) pattern factor — so far-ambiguous-pulse attenuation from the elevation plane is not modeled; the principal effect of that simplification falls on the absolute ambiguity level, and it is registered with the model caveats (Section 7.3).

Admission uses a *discrete* worst-case readout, *not continuous* evaluation: the AASR of a task is evaluated at 11 equally spaced *in-window* samples and the maximum must not exceed the registered $-13$ dB admission threshold at generation time. Feasibility statements in this paper inherit that caliber — a

selection is feasible with respect to the generator's 11-point admission rule, and the evaluator performs no AASR recheck of its own.

### *3.4. Constraints and objectives*

The decision vector of a scenario with $N$ tasks is $x = (s, \tau) \in \{0,1\}^N \times [0,1]^N$: a selection bit per task and a start-time fraction per selected task. The imaging mode $m_i$ is *not* a decision gene: the pair (required resolution, scene size) is mapped to a mode by the library table, a *system design input with product order semantics* — the platform is told what product a customer ordered — and the mode assignment is frozen for the data of this paper (Section 3.5). What the scheduler does inherit from the order is the mode's parameterisation: for sliding spotlight and spotlight the design parameters are functions of $\rho_{\text{req}}$ (constraint C5), so the order reaches the dwell through the mode rather than through a decision variable. Joint mode-and-time allocation is future work.

Five constraint families apply.

- **C1 (geometry, generation side).** A selected start must fall inside a visibility window of the task whose node satisfies the serviceability evidence: *every* node in the dense domain carries *at least one* in-band transit over the span (25/25 nodes over 72 h, per-node reading taken on the registered domain construction).

- **C2 (inter-task transfer).** For consecutive tasks in start-time order, the gap must cover $(1 + 0.25)\, T_{\text{dwell},i} + t_{\text{man}}(d_\psi, \omega_{\text{start}}, \omega_{\text{end}}) + t_{\text{settle}}$: the 25% pad multiplies the *dwell* (Kim's own margin semantics, stated in four places in Kim, Hongrae and Park, Joonil and Chang, Young-Keun and Lee, Soo-Ho 16), never the maneuver, and the settle term is an unpadded measured constant ($t_{\text{settle}} = 2.0$ s). Production evaluates $t_{\text{man}}$ as a registered hybrid: the analytic model stands alone only where it reaches at least half of the measured calibration anchor (1.84 s, Kim scenario 1); below that trigger the per-transition maneuver bound falls back to the worst measured required maneuver time (4.07 s, pure maneuver, Kim scenario 2)

— a conservative substitution that raises a visible warning in the implementation. The conflict predicate uses the bare dwell: pads are scheduling margin, not physical occupancy.

- **C3 (energy/duty).** Per-task duty is normalised against the mode-library envelope reference $T_{\mathrm{env,max}} = 10.5830$ s (the largest feasible scene dwell at the envelope-edge squint, attained by SPT) with mode weights STR/SLS/SPT = 1.0/1.15/1.3; the normalisation denominator is the largest weight times the envelope reference, $1.3\,T_{\mathrm{env,max}}$ (the C4 reference below omits this weight factor).

- **C4 (solid-state memory).** Per-echo bytes follow PRF$\cdot T \cdot N_r \cdot b_q \cdot 2/8$ with $N_r = 10\,000$ range samples and $b_q = 3$ bits per component (TerraSAR-X-style block-adaptive quantisation), normalised against the same per-window reference as C3 (each measured window's own slant range, the caliber registered for this scenario world) under the single production PRF of 3630 Hz.

- **C5 (order resolution, evaluation side).** Every selected task must be imaged at its ordered azimuth resolution $\rho_{\mathrm{req}}$. For sliding spotlight and spotlight the mode parameters are derived from the order at the selected start's squint $\psi_{\mathrm{sel}}$ — $A_{\mathrm{task}} = 2\rho_{\mathrm{req}} \cos\psi_{\mathrm{sel}}/D$ and $\beta_{\mathrm{task}} = \lambda \sec\psi_{\mathrm{sel}}/(2\rho_{\mathrm{req}})$ — so the delivered resolution equals the order by construction; the identity is re-verified at every dwell evaluation against a $10^{-12}$ tolerance, and a violation raises instead of being scored. The claim is pointwise at that start: inside the imaging interval the delivered resolution follows $\rho_{\mathrm{req}} \cos\psi_{\mathrm{sel}}/\cos\psi(t)$, so this paper does not claim a constant delivered resolution across a whole imaging pass. The stripmap mode (STR) delivers the library floor of 2.4669 m: it has no task-level parameter, so an order finer than the floor has no admissible mode for that task, and a tighter order there is a step at the floor rather than a continuous degradation.

Two structural facts about C3/C4 are disclosed as design semantics rather

than discovered: both budgets are *structurally never binding* inside the squint envelope (their normalised values cannot exceed 1.0 there, so the binding channels are C2 and interval conflicts), and the task priorities that drive the objective are drawn uniformly at random as randint$(1, 10)$, i.e. they carry *no physical* provenance — they are demand-side bookkeeping, not measured value.

The problem maximises the served priority fraction $f_1 = \sum_{i \in S} p_i / \sum_i p_i$ over the selections $S$ that satisfy C1–C5. Coverage is the only objective. The quality index of Section 3.3 carries no selection pressure: the search optimises one criterion rather than two, a front is a tie-set on coverage, and the index appears only as the fulfillment reading. The search method is described next.

A reader who meets this model through earlier versions of this line will expect a quality aggregate to be traded against served priority, so the change is stated explicitly. This is not a re-parameterisation of the same problem: the search returns a tie-set instead of a front, and the quality index moves from a decision criterion to a measured property of whatever the scheduler serves. The change is registered as an amendment to the project's preregistration, which vacates the quality-headroom estimand and registers the coverage–demand curve of Section 3.5 as the main result form.

### *3.5. The demand-tightening axis*

The experimental axis of this paper is external to the scheduler. A scalar $s$ multiplies every task's ordered resolution, $\rho_{\mathrm{req},i}(s) = s\,\rho_i^{\mathrm{order}}$, with $s = 1$ reproducing the order as placed and $s < 1$ a tighter order; nine values are frozen, $s \in \{0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 2.0\}$. Scaling touches the order only: the scenario population — geometry, task set, and visibility windows — is not regenerated, so every point of the grid is one scenario world seen through a different order.

The axis is swept under a frozen mode assignment. Which mode a task receives is not re-derived as the order tightens: the assignment is the one the scenario world was generated with, inherited unchanged (the grandfather decision registered in formula-spec §8.17-9). The reason is comparability — re-

cutting the mode mix at every grid point would confound the coverage response with a routing change — and it fixes the scope of every conclusion that follows. The results are order sensitivity under a fixed mode assignment, not a general statement about how coverage trades against resolution when a router is free to move tasks between modes; the experiment that would answer the general question is a different one and is not run here.

The mode assignment matters most on the stripmap family, because that is where the floor sits. A family whose tasks are all stripmap cannot respond to a tighter order continuously: the delivered resolution is the antenna floor, so the family either serves at the floor or finds no admissible mode. The single-orbit family R1C meets that floor inside the frozen grid — at $s = 0.8$ the family has no feasible individual in 10 of 10 seeds — and that outcome is an infeasibility record, not a measured coverage of zero. The two statements are kept apart everywhere in this paper: an infeasible grid point is not a point on the curve, and it is never plotted, averaged, or quoted as one.

Three different mode rules are in play in this project, and a reader should not mistake one for another. (i) The admission path routes an order to the coarsest library tier whose zero-squint resolution meets it (`aasr.py:178-189`); that rule consults the requirement and the library alone, one tier per task, with no scene-size test. (ii) The registered selection policy adds the feasibility test ($A_{\text{task}} \in (0, 1)$, the mode envelopes, the spotlight footprint bound, the ambiguity gate) and a two-key rule on scene size — spotlight inside the footprint, sliding spotlight beyond it, stripmap at the coarse end — and is the policy registered for narrative and for future families (formula-spec §8.17-9). (iii) The data of this paper follow neither of those at run time: they carry the grandfathered assignment described above. Where this paper says that the mode follows the order, it means that the parameters of the *assigned* mode follow the order, not that the assignment itself changes.

Coverage is measured on this axis as a reading: for each family and each grid point, the median over ten seeds of the served priority fraction of the best-covered member, at a single search budget. Three shape criteria are frozen

against the curve — monotone non-decreasing medians across the grid, a saturation arm ($s = 2.0$ at least $s = 1.5$), and a tight arm ($s = 0.8$ below $s = 1.0$) — and they are inherited unchanged from their earlier registration rather than tuned to this world's data. Section 6 reports them as measured, failures included. Coverage admits no pass/fail verdict here: every unit's front is a tie-set carrying one coverage value, so there is nothing for a verdict to decide, and the curve is reported as a reading.

## 4. Solving Approach

In scheduling-theory terms the problem of Section 3 is a single-machine scheduling problem with time windows and sequence-dependent setup times: the satellite is one resource, each task carries visibility windows, and the inter-task transfer of constraint C2 charges a maneuver time that depends on the attitude endpoints of the consecutive pair — a sequence-dependent setup in the classical sense. The imaging dwell adds a second classical construct: because the dwell follows from the mode and the squint of the selected start, a task's processing time is not a constant but a function of where it is started within its window, a position- and time-dependent processing time of the kind studied in the time-dependent scheduling literature [5, 4, 10]. No complexity claim is made in this paper: no hardness proof or approximation result is attempted, and the evolutionary search of this section is offered as a measurement instrument on a registered scenario world, not as a contribution to scheduling theory. Bounding methods for the fixed-dwell agile-observation problem exist [8]; how they transfer when the processing time is endogenous is an open question this paper does not address.

### *4.1. Encoding and evaluation*

The solver is a single-objective evolutionary search over the chromosome of Section 3.4: a selection bit and a start fraction $\tau \in [0, 1]$ per task — no mode gene, since the mode is fixed by the product order — so the densest real-window

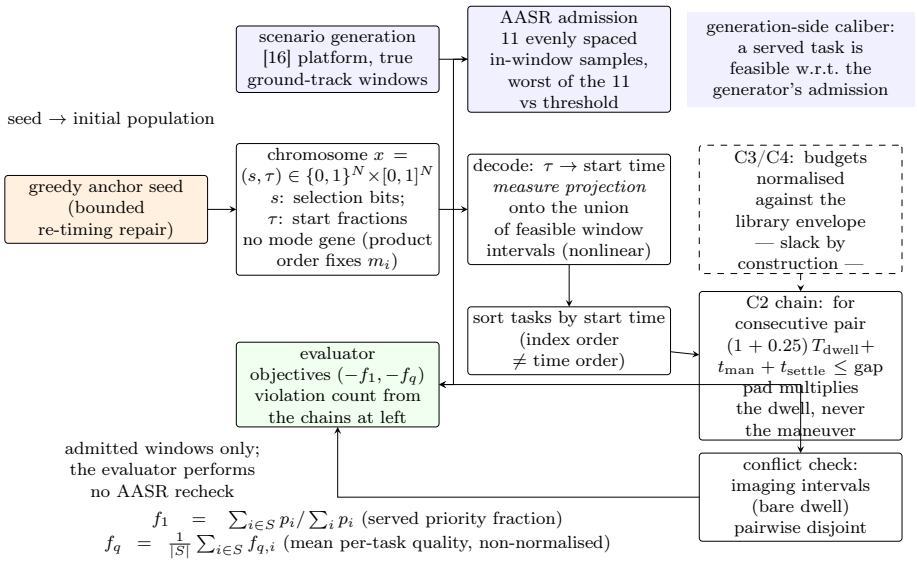



Figure 2: The scheduler's encoding-evaluation loop. Generation side (blue): scenario construction and AASR admission (11 evenly spaced in-window samples, worst-of-11 against the threshold). Genome: selection bits and start fractions, no mode gene; the greedy anchor seed is injected into the initial population. Decode is a measure projection of $\tau$ onto the union of feasible window intervals (nonlinear by construction). Evaluation: tasks are sorted by start time (index order is not time order), the C2 chain charges the $(1 + 0.25)$ pad on the dwell — never on the maneuver — plus the maneuver and settle terms, conflicts use the bare dwell, and C3/C4 (dashed) are slack by construction inside the envelope. The evaluator itself performs no AASR recheck: feasibility is relative to the generator's admission caliber.

family (R3H, defined with the scenario families in Section 5.2) encodes the 99 admitted tasks (of 100 drawn) over 753 enumerated windows as $2 \times 99 = 198$ decision variables; at the coarsest end the control family (R1C, 20 drawn and 20 admitted) encodes 40. Decoding projects $\tau$ onto the union of feasible window intervals of the admitted windows: a served task is feasible with respect to the generator's discrete 11-point admission rule (Section 5.1), and the evaluator performs no AASR recheck of its own; feasibility statements in the results are relative to that generation-side caliber, not to an independent physical guarantee (Fig. 2).

*4.2. Anchor-seeded initialization*

The initial population contains the greedy anchor: a bounded single-pass re-timing repair of the all-selected schedule. The seed is what makes the $f_1$ normalisation meaningful — every front carries a solution anchored at the served-priority reference — and its injection into the population is checked by a dedicated cross-check that once exposed spurious violations from time-inverted index pairs, which is also why the evaluation loop sorts by start time before charging the C2 chain.

*4.3. Search operators and selection*

Variation applies simulated binary crossover ($\eta = 15$, per-variable probability 0.9) and polynomial mutation ($\eta = 20$) uniformly over the real-valued chromosome — the selection bits are ordinary real genes decoded by a 0.5 threshold, so no separate binary operator acts on them. Because the problem carries one criterion, the non-dominated sorting of the standard machinery ranks by that single value alone [15], and the survivor set of a unit is a tie-set on coverage rather than a trade-off front: the search returns the best-covered selections it reaches and the tie-set records their common coverage (Section 3.3). The search runs under a fixed random seed per unit, single-threaded, with manifest task-list order everywhere (no set or dictionary iteration order may influence a run), so every unit is bit-reproducible.

*4.4. The measure normalisation of $\tau$ and its implicit bias*

One parameterisation choice is disclosed as intentional. A start fraction $\tau$ indexes position within the *feasible measure* of a task's window union, not within wall-clock time: decoding is the measure projection of Fig. 2. Because the projection is measure-normalised, a uniform grid on $\tau$ samples wider windows more finely — the parameterisation carries an implicit weighting toward wider windows. This bias is registered as a deliberate design decision: it is not part of the main specification or its preregistration, and its registration is carried in the project's dated design-amendment record (review note of architecture decision

AD-3). No selection mechanism in the loop undoes the weighting; it is part of the search design, and every reading in Section 6 inherits it.

### *4.5. The search face and the construction face*

The response this paper reports has two faces, a search face and a construction face, and the separation between them is measured rather than assumed. Over the 260 units that produced an artifact (nine demand points × three families × ten seeds, less the ten exit-11 units of Section 6.2), the greedy anchor already attains the unit's coverage maximum in 258; the search exceeds its own anchor in exactly two units — the control family R1C at the order point, seeds 100 and 105 — by +0.0427 and +0.0463 of the served priority fraction. That is a statement about where the coverage response comes from, not a statement about solver quality. The chromosome can do only two things: move a start inside the feasible measure of its window union, and drop tasks from the selection. Which tasks remain serviceable at a given order is therefore decided before the search runs, by the mode assignment and by the parameter adaptation that carries the order into the dwell; the search's job is to reach the level that admission sets. Within this world the anchor is what reaches it, and the two units where the anchor does not are recorded rather than smoothed. The consequence for reading Section 6 is direct: the coverage differences between demand points are a property of the construction, and no statement in this paper attributes them to the optimiser.

## 5. Experimental Design and Setup

This section describes the scenario world, the family and caliber system, and the preregistered test design.

### *5.1. Scenario construction*

The scenario world is built from *real ground-track transit enumeration*, not from an arbitrary orbit phase point: visibility windows come from enumerating true ground-track passes over the scenario span, with a 15 s scan step, gap

merging at 300 s (a registered band parameter, invariant across the registered 150/600/1200 s sensitivity band, not a physical lower bound), and 33-point edge refinement. The registry's own ledger records why: under the earlier phase-point pass rule only 0 of 8 clusters were serviceable after the incidence fix, against 6 of 8 under true enumeration. The demand mix itself — resolution requirements and scene sizes — is a registered synthetic template parameter, not a measured campaign profile; the priorities that drive the objective are uniform random integers with no physical provenance (Section 3.4), so the mode composition of the served set reflects the template's requirement distribution.

Orbit propagation is two-body only — no $J_2$ or atmospheric terms — so the registered repeat figure is the *simulated* ground-track repeat of 14.97 orbits $\approx$ 23.93 h (one sidereal day over the two-body period); the figure that includes $J_2$ nodal drift describes the physical orbit and is deliberately not the one this simulator produces. The surface normal used in incidence computations is an *approximate* normal: its maximum deviation from the true ellipsoidal normal is 0.125° (at $\approx$30° latitude; 0.109° at the dense-domain centre), with no effect on band-boundary judgements at the registered margins. The dense domain is a 1000 km square centred at 20°N/0°E, a $5 \times 5$ node grid at 250 km spacing with per-cell target jitter, a 3-day span in coordinated universal time (UTC). The serviceability criterion is per-node: every node carries at least one in-band transit within the span, measured 25/25; the literal best-incidence reading of the dense-domain constraint fails at the centre node (near-nadir passes exist there but yield no in-band window), which is why the operative criterion is the per-node one. The 25/25 reading is a construction-time measurement on the full-span domain; the manifest ledger stores aggregate admission counts, not per-node columns.

The AASR admission criterion is evaluated per family, and the two families differ by design — a discrepancy we disclose rather than harmonise post hoc. Synthetic M-family windows are admitted by a window-edge single-point read-out (the AASR at the window edge squint only), whereas the real-window R families sample the AASR at 11 evenly spaced in-window instants (lo$+$(hi$-$lo)$\,j/10$,

$j = 0, \ldots, 10$) and admit the window only if the worst of the 11 samples meets the threshold. A sample outside the physical sliding domain contributes an exclusion, so it can never mask an unserviceable window. The window-edge single-point rule *underestimates* the in-window worst case by about 4 dB, so the two admission rules are stated per family wherever admission is discussed; the R-family rule is the discrete in-window worst-case caliber that all feasibility statements in this paper inherit (Section 3.3). This sampling is *discrete*, *not continuous* evaluation: a window admitted under it is feasible with respect to the 11-point rule, and the evaluator performs no AASR recheck of its own. Two further conservative narrowings bound the admissible window set and are carried with the same caliber: a pass is not admitted when the scene along-track footprint exceeds the beam-limited bound $R_0\,\theta_a$ at the closest-approach slant range of that pass (binding for SPT), and the envelope mask admits starts only inside a 1% inward guard of the mode envelope, within a $\pm 120$ s span around the pass closest approach, with each admitted mask run inset by one scan step ($\approx$1 s) per end.

### *5.2. Families and calibers*

Three real-window families form the experiment matrix, and their sizes are stated in the two numbers the registered density ladder fixes, the tasks drawn and the tasks admitted (serviceable): R1C (20 tasks drawn, 20 admitted, 158 windows) is the single-orbit control; R2F (50 drawn, 46 admitted, 342 windows) and R3H (100 drawn, 99 admitted, 753 windows) are the dense-cluster treatments. Task counts follow the registered density ladder $N = 20/50/100$; R4H ($N = 500$) was retired at registration because the single-transit STR pass-count scaling exceeds the registered span at $N \geq 150$ (the retirement rationale is registered; no capacity number is asserted anywhere in this paper, since the serial bound has not been recomputed under the present real-window scenario world).

The mode composition of each family is inherited from the grandfathered assignment of Section 3.5 rather than re-derived at run time: every R1C task is stripmap, R2F routes its tasks between sliding spotlight and spotlight and

carries no stripmap task, and R3H mixes all three modes. That composition sets the scale of the quality index a family can reach (Section 3.3), but it is a property of the demand template that generated the scenario rather than of any schedule, and no reading in this paper compares index levels across families. The control family's served fraction saturates at 1.0 from the order point upwards; the saturation is anchor-construction-carried (Section 6.5) and is read descriptively, never as a performance result.

The artifacts are the project's run-lineage node L8 (caliber v5), and they are not a refresh of the previous world but an explicit re-judgement. Ticket 49 made the sliding-spotlight and spotlight parameters follow the order per task (constraint C5), so the delivered resolution at the selected start equals the order by construction; that changes what is admissible at the data level rather than merely how a reading is labelled, and the earlier artifacts therefore no longer describe the same model. L8 re-runs the sweep over the same frozen scenario population — the task set, the visibility windows and the mode assignment are carried over unchanged, so the demand axis is the only thing that moves — at a single registered budget of 40/60 population/generations. The project's run-lineage register records the L8 node with its parent, its frozen head and its adjudication, and every reading in Section 6 is recomputed from those artifacts by the manuscript-number checker before the section is read as a result.

The R-family $f_q$ is *not normalised*: the per-window slant-range geometry expands the units scene-dependently, so index levels are incomparable across families and are never pooled into a single span or compared across families. The synthetic M families remain available as a controlled reference; their one still-valid artifact is annotated, wherever cited, as a version-2-caliber controlled reference whose output provenance is unregistered, and no M-family number is mixed with R-family numbers in this paper.

### *5.3. The demand scan and its frozen criteria*

*The scan..* The experiment is an external scan rather than a hypothesis test about a control: a scalar $s$ multiplies every task's ordered resolution (Sec-

tion 3.5), nine values of $s$ are frozen, $s = 1.0$ reproduces the order as placed (the order point), and each grid point is read as the median over ten seeds of the served priority fraction of the best-covered member, at the registered 40/60 budget. The scenario population is not regenerated between points, so the nine readings are nine views of one world through nine orders.

*Three criteria, frozen before this world's data were read..* Three shape criteria are registered against the curve and are inherited unchanged from their earlier registration rather than tuned to the data. C1 asks that the medians be monotone non-decreasing across the grid; C2 is the saturation arm, the median at $s = 2.0$ at least the median at $s = 1.5$; C3 is the tight arm, the median at $s = 0.8$ below the median at $s = 1.0$. A cell whose ten seeds all exit without a feasible individual enters the criterion evaluation as a zero-coverage median, and it is the tight arm (C3) and the monotone arm (C1) that read that cell; the saturation arm (C2) compares the two loosest points and does not. The zero-coverage entry is a recorded outcome rather than missing data, and the reporting discipline of Section 6.2 keeps such a point off the coverage curve and never averages it into one. They are the registered content of the preregistration amendment that also vacated the quality-headroom estimand of the earlier line and registered the coverage–demand curve as the main result form (Section 3.4). Their verdicts are reported as measured, failures included, and the criteria are not re-tuned after the fact: a failed criterion is a result about this world, not a defect in the criterion to be edited away.

*What the amendment retired..* The earlier design carried an effect size, a power calculation and a power curve, all of them calibrated on the quality headroom of a front over its anchor. That estimand does not exist in this model — a unit's front is a tie-set on coverage, so there is no quality quantity to be gained — and the power figure is retired with it rather than reproduced: a power curve about an estimand this paper does not report would be a number in search of a claim. Nothing in Section 6 depends on it.

*One budget, stated..* Every point of the grid is read at the single registered budget of 40/60; no budget ladder is run in this world, and no frontier-richness statement in either direction is made from these artifacts.

## 6. Results and Analysis

All readings in this section are taken from the L8 artifacts at caliber v5 (Section 5.2) and are cross-checked against the reading ledger that stores each of them with its unit, caliber label and source path. The grid is the nine-point demand scan of Section 5.3, every point read at the registered 40/60 budget.

### *6.1. The coverage–demand curves*

The result of the paper is the coverage–demand curve (Fig. 3): the served priority fraction of the best-covered member, median over ten seeds, as a function of the demand-tightening factor. Three families produce three different shapes, and the difference between the shapes is the result rather than a nuisance variation to be averaged away.

The 100-drawn / 99-admitted family is the one the tightening actually reaches. Its median coverage falls monotonically with the tightening: 0.4827 at $s = 0.8$, 0.6884 at $s = 0.9$, 0.9485 at the order point $s = 1.0$, then 0.9766 at $s = 1.1$, 0.9812 at $s = 1.2$, 0.9882 at $s = 1.3$, 0.9882 at $s = 1.4$, 0.9900 at $s = 1.5$ and 0.9900 at $s = 2.0$ (family R3H). Tightening the order from the order point to the tightest point that still has a feasible individual costs about half of the served priority mass; loosening it from the order point to the coarsest point buys back under five points. Most of the response therefore sits on the tight side of the order point, which is the side an operator would actually consider.

The 50-drawn / 46-admitted family is nearly flat and sits high. Its median coverage reads 0.9764 at $s = 0.8$, 0.9737 at $s = 0.9$, 0.9764 at $s = 1.0$, 0.9764 at $s = 1.1$, 0.9960 at $s = 1.2$, 0.9960 at $s = 1.3$, 0.9924 at $s = 1.4$, 0.9924 at $s = 1.5$ and 1.0000 at $s = 2.0$ (family R2F). On this family a tighter order is close to free in coverage: the two local declines, of 0.268 and 0.359 percentage

points, are small enough that they are given their own accounting (Section 6.4) rather than read as a trend.

The control family is the single-orbit one, 20 drawn and 20 admitted, and its response is a step rather than a curve. Its median coverage reads 0.5398 at $s = 0.9$ and 1.0000 from $s = 1.0$ all the way to $s = 2.0$ (family R1C): the family either serves the whole priority mass at the stripmap floor or serves a strict subset of it, and there is no intermediate regime for it to occupy. Its tightest grid point behaves differently again and is kept apart from the curve (Section 6.2).

Read together, the three curves say what the mode assignment decides. A family whose tasks are all stripmap has a floor and therefore a step; a family that mixes sliding spotlight and spotlight absorbs a tightening in its parameters and pays almost nothing in coverage; and the family that mixes all three modes, at the largest task count, is the one on which the tightening bites. That ordering follows the construction of Section 4.5 — the modes and their parameter adaptation decide which tasks stay serviceable — and it is not a property of the search.

### *6.2. The order point, the floor step, and the cost of a tighter order*

Every cost in this subsection is measured against the family's own saturation point — the first grid point at which its median reaches the family's maximum — and never against another family or another world. That choice is what makes each number comparable to nothing but itself, and it is stated on every line that carries one.

R3H pays a coverage cost of 4.15 percentage points at the order point against its own saturation at $s = 1.5$, the first point at which its median reaches 0.9900. R2F pays a coverage cost of 2.36 percentage points at the order point against its own saturation at $s = 2.0$, where its median first reaches 1.0000. R1C's cost against its own saturation is 0.00 percentage points, and that zero is a statement about where the family's saturation sits — the order point is its saturation point — not a claim that a tightening is free for every family.

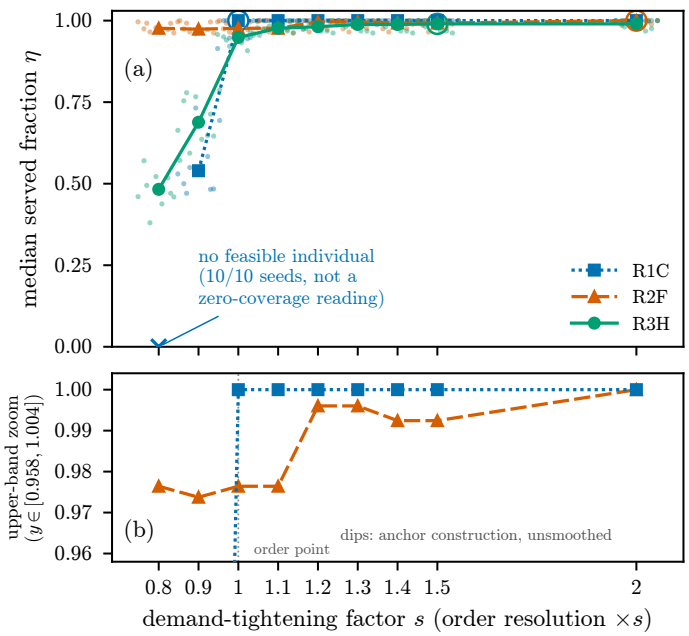



Figure 3: Coverage versus the demand-tightening factor $s$ for the three real-window families (L8 world, caliber v5, single registered budget 40/60, ten seeds per point). (a) Median served priority fraction of the best-covered member, with the individual per-seed readings drawn as dots; a hollow marker sits on the first grid point at which a family reaches its own maximum. The cross at $s = 0.8$ of the control family is *not* a zero-coverage reading: all ten seeds of that cell exited with no feasible individual (Section 6.2). (b) Upper-band zoom, $y \in [0.958, 1.004]$, where the two high families and their small local declines are legible; the truncated range is stated in the axis label. The dotted vertical line in panel (b) marks the order point $s = 1.0$.

The control family's tightest grid point is a different kind of fact and is separated from the curve on purpose. R1C at $s = 0.8$ has no feasible individual in 10 of 10 seeds: every unit of that cell exited with the no-feasible-individual code, so the family admits no admissible schedule at that order under the frozen mode assignment. That is not a coverage of zero, and the two must never be interchanged. A zero would say that the search considered schedules and served none; what happened is that the order at $s = 0.8$ asks for a resolution finer than the stripmap floor the family is fixed to, so there is no admissible schedule to consider. The figure draws the point as a cross against the axis carrying that

label, the ledger stores it as a status rather than as a value, and no average, no smoothing and no line segment passes through it. Every statement about this point in the rest of the paper keeps the same distinction.

### *6.3. The frozen criteria and their verdicts*

Table 1 reports the three criteria of Section 5.3 as measured. The control family and the large family meet all three; the control family's infeasible tight point enters the criteria as the zero-coverage outcome registered in Section 5.3. The control family's tight-arm verdict — met — is a direct consequence of that convention: under an exclusion convention the tight cell would be undefined and the verdict would carry no value; the monotone verdict (C1) is unchanged under either convention. The 50-task family meets the saturation arm and fails the other two: its medians are not monotone across the grid (C1), and its tight arm does not fall (C3). Both failures are reported as results about this world.

C3's failure carries information about the model, and it is the falsification of a reading rather than a null result. In the previous world this family's tight arm carried a large penalty — the median at the tightest point was 0.446 of the served priority mass — and that penalty is *withdrawn*: under the caliber-v5 model the same family reads 0.9764 at $s = 0.8$, above its own order-point value, so the tight arm no longer falls at all. The withdrawn value is a historical comparison and is labelled as such wherever it appears; it is not a reading of this world and it is not an estimate this paper stands behind.

What replaces it is smaller and better localized. The tightening still costs this family something, but the cost is measured at the order point against the family's own saturation (2.36 percentage points, Section 6.2), not at the tight arm. The mechanism is the parameter adaptation of constraint C5: the mode parameters are derived from the order at the selected start, so the delivered resolution equals the order by construction, and a tighter order therefore spends dwell rather than serviceability. The family pays in imaging time; it does not pay by losing tasks, until the dwell term grows enough to collide with the transfer constraint. Saying that the cost has vanished would be wrong on both halves

Table 1: The three shape criteria of Section 5.3, as measured on the nine-point demand grid (L8 world, caliber v5, single registered budget 40/60, ten seeds per point). The criteria were frozen before this world's data were read and are not re-tuned after their verdicts.

| family (drawn / admitted) | C1 monotone | C2 saturation | C3 tight arm |
|---|---|---|---|
| R1C (20 / 20) | met | met | met |
| R2F (50 / 46) | not met | met | not met |
| R3H (100 / 99) | met | met | met |

of that sentence: the tight-arm penalty is gone, and the order-point cost is not.

C1's failure is the pair of local declines that Section 6.4 accounts for, and it is not removed by re-tuning. A criterion edited after its verdict is not a criterion: both failures stand as this world's answer to questions that were fixed before its data were read.

*6.4. The two local declines: an anchor-construction effect*

Two local declines interrupt the flat band of the 50-task family: 0.268 percentage points between $s = 0.8$ and $s = 0.9$, and 0.359 percentage points between $s = 1.3$ and $s = 1.4$. Both are attribution results, and the attribution is not the one a reader would expect.

Neither is a search miss. Over all 180 sweep units of the two treatment families the best-covered member of a unit's tie-set equals the greedy anchor's value exactly, and the search-only maximum is empty; across the whole 260-unit world the search exceeds its anchor in the two control units of Section 4.5 and nowhere else. A decline produced by a search that never misses is not a search miss. It is produced by the anchor construction itself, which is deterministic and reproducible: the anchor re-times the all-selected schedule in a single bounded pass, and at a given order that pass can serve one task fewer or one task more than at the neighbouring order.

The per-seed accounting makes the mechanism visible. Between $s = 0.8$ and $s = 0.9$ two seeds move in opposite directions — one loses 1.83 percentage points and one gains 0.42 — and the median falls because the net of those two

displacements is negative. Between $s = 1.3$ and $s = 1.4$ the same shape recurs with different seeds: one loses 1.14 points and one gains 0.42. In both cases the decline is the median of ten readings following two of its members, not a property of the demand axis.

This is a different object from the search-artifact verdict of the earlier line, and the difference matters for how the reading is used. That verdict retired a reading because a larger search budget removed it: what was retired was an artifact of the search. Here no budget changes the picture, because the effect sits in the construction, which does not depend on the search at all. And nothing physical follows from either decline: two opposite single-seed displacements are not a monotonicity claim in either direction, and no statement about the demand axis is made from a movement in the third decimal place.

*6.5. Demand satisfaction and retention*

The curve says how much priority mass is served; a second reading says what that service amounts to. Over the nine demand points the tie set serves 477,308 task instances, distributed over 9,430 tie-set records: a served task instance is one tie-set member's selection of one task, not a distinct task, and the count is the sum over every record of the tasks that record selects. The delivery identity that constraint C5 enforces — delivered resolution equal to the order at the selected start — holds by construction and is checked by a raise rather than by a score, so no artifact stores a per-record verdict; the observable is the constraint violation, whose maximum over every record is 0.0, and no violation is recorded relative to the evaluator for any unit of the sweep. Ten of the 270 launched units exited with no feasible individual (the control family at $s = 0.8$) and are excluded from the served count by the driver ledger rather than folded into it as zeros.

The retention reading measures the same service from the schedule's side, and it travels with a caliber that must travel with it. On the STR-clip-only caliber — the feasible measure retained after the stripmap clip — retention reads 51.14% for the control family, 100.00% for the 50-task family and 71.00%

for the 100-task family. The earlier line's retention figures, which were computed under the retired SLS/SPT narrowing, are superseded and are not quoted here.

### *6.6. Reported readings: fulfillment, its relation to coverage, and the delivered variant*

Three readings are reported beside the curve. None of them is an optimization target, and each one travels with the caliber label that defines the quantity it states.

*The fulfillment ratio..* For a served task the ratio is its $\mathrm{f}_q$ at the realised window and the selected start's squint, over the best $\mathrm{f}_q$ the task could reach over all of its windows at zero squint, so the ratio is at most one; it is averaged over all tie-set members and reported, never optimised. At the order point the fulfillment ratio reads 0.6580 for R1C, 0.7764 for R2F and 0.7447 for R3H. Its label travels with every number it states: *[library caliber $f_q$ | per-task $f_q$ at the realised window and squint over the zero-squint optimum over all windows | mean over all tie-set members | reported reading, not an optimization target]*. What the reading says is narrow and is kept narrow: it measures how much of the zero-squint optimum over a task's own windows survives the choice of window and start, not how good the image is by any calibrated radiometric standard (Section 7.3).

*Its relation to coverage..* No directional claim is registered between coverage and the fulfillment ratio, and none is made here. What is reported is the coefficient and its sample size, with the caliber named on the same line: the Pearson coefficient between the family-by-grid-point coverage and fulfillment readings is $r = 0.329$ over $n = 26$ points, and over the pooled units it is $r = 0.249$ over $n = 260$. These are relation readings of this world's data; the sign is not a property of the mode continuum, and no mechanism is claimed for it.

*The delivered variant, and the platform capability behind it..* The descriptive delivered variant $f_{Q,\mathrm{deliv}}$ of Section 3.3 is reported once, under its own label: *[delivered caliber $f_{Q,\mathrm{deliv}}$ | descriptive variant, not an optimization target | not*

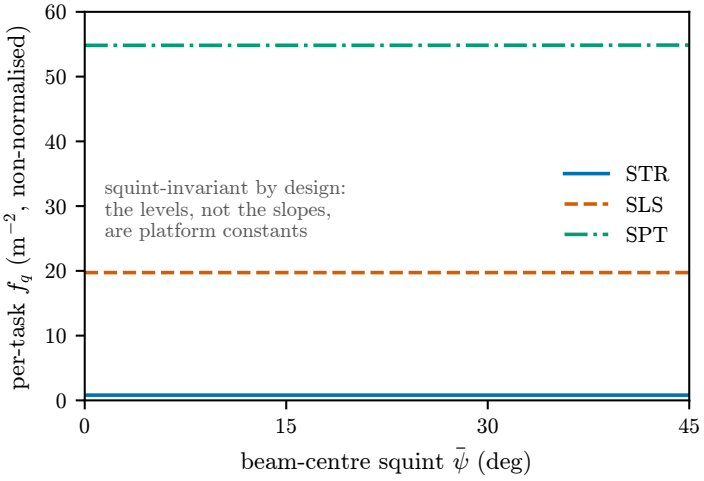



Figure 4: The per-mode quality index versus beam-centre squint for the three library modes, evaluated through the frozen platform fixture at the zero-squint library caliber (non-normalised, $m^{-2}$). The levels are flat by design: the per-mode constants of Section 3.3 differ by orders of magnitude while the SPT drift across the full envelope is +0.067%. The figure states a platform capability, not a result of any schedule.

*comparable with the library caliber | no stripmap domain].* At the order point its median for R3H (100 drawn / 99 admitted) is 37.62 m$^{-2}$, and the median squint-loss factor on the same cell is 1.3286. This is a descriptive variant, not comparable with the library caliber, and the two are never placed in the same table. What the platform fixes is the capability statement of Section 3.3: the per-mode index levels are separated by orders of magnitude and are flat in squint (Fig. 4), so which mode a task receives dominates the value any plan can reach on this axis, and the scheduler moves a plan inside the range the demand template sets rather than setting the range itself.

## 7. Discussion and Limitations

### *7.1. Scope and negative claims*

Five scope statements are registered before anything is claimed. First, the capacity law carries a per-pass caliber: the registered serial bound is defined

per pass, not per orbit, and the closure (the served fraction of the bound) is a containment reading — it says the schedule fits inside the bound, not that the bound validates the scheduler. The minimum-over-pairs transfer time that enters it is a legal lower bound on the true transfer, never an average or a worst case. Under the present real-window scenario world the serial bound has not been recomputed, so this paper reports no capacity number in any form; the law, its caliber, and its discipline are the contribution, and the recomputation is future work.

Second, the main claim is bounded by its mode assignment. The results of Section 6 are order sensitivity under the frozen mode assignment of Section 3.5: the curve answers what tightening the order costs when the mode a task receives does not change. It does not answer how coverage trades against resolution when a router is free to move tasks between modes, and that experiment is not run here. Third, the curve is read at one registered search budget (Section 5.3), so it is a reading of this world at that budget and not a statement about a converged frontier in either direction.

Fourth, the trade-off this study claims is the resolution-versus-dwell-time axis of the mode continuum. The antenna-design question of resolution against swath is a sensor-layer consequence that requires a width-versus-factor model this simulation does not have; it is reported as a limitation and future work, not as a demonstrated result. Fifth, the operational layer: the model covers single-pass scheduling decisions only — no dynamic task arrival, no downlink or ground-station modeling, no cloud-cover attrition, and no replanning loop — and no wall-clock runtime figure is registered for the promoted runs, so the platform is offered as a falsifiable scheduling substrate, not as an operationally timed planner; operational-cost evaluation is future work (Section 8).

### *7.2. Azimuth ambiguity versus the literature: a direction contrast*

The registered correction behind this section is blunt: in this model the ambiguity-versus-squint direction is the *opposite* of the literature anchor it would most naturally be cited against. Under the production PRF and the registered

calibration grid, the sliding-spotlight ambiguity ratio improves monotonically with squint: from $-21.35$ dB at zero squint to $-23.32$ dB at the $26^\circ$ calibration-grid edge (the registered SLS cap is $26.9^\circ$). The stripmap ratio likewise improves ($-21.35$ to $-23.32$ dB over the same registered 0–$26^\circ$ calibration span), and the spotlight ratio is squint-invariant at $-7.05$ dB, exactly as the degenerate-integral structure of Section 3.3 predicts. Song et al.'s platform, by contrast, degrades with scan angle (the anchors quoted in Section 2) [22]; those values are cited as that platform's literature readings only. The contrast also carries its model-dependence: it is a consequence of adopting the rotation-centre semantics (under which the compression factor is squint-independent) together with the registered omission of the absolute Doppler centre, and no experiment in this paper discriminates the two steering semantics; the literature-side mechanism — a degrading compression factor competing with bandwidth narrowing — is not modeled here, so the comparison establishes a difference of registered model behavior, not a measurement that refutes the anchor.

The mechanism is the Doppler bandwidth. In this model the spectral compression factor is the design constant $a_{\mathrm{res}} \equiv A$ at every squint (Section 3.1), so the *only* squint-dependent channel into the ambiguity integral is the Doppler bandwidth $B_D = v_s \cos\bar{\psi}/\rho_a(m, 0)$: squint narrows $B_D$, a narrower band aliases less of the ambiguity power, and the ratio improves. The registered direction contrast therefore differs by mechanism, not merely by sign: it is what remains once the pseudo-boundary of the linearised steering law — under which the compression factor itself would collapse with squint — is retired as the extrapolation artifact it is (Section 3.1). We flag the contrast rather than reconcile it: the two statements are made under different steering semantics, and the comparison is only legitimate with that stated. A caliber note on the spotlight reading: $-7.05$ dB is the library-edge constant evaluated at the platform-level scene extent (5 km, worst edge), whereas admission evaluates each task at its own scene extent through the 11-point in-window rule of Section 3.3: the spotlight aliasing term scales with scene size over range. The two figures are therefore different calibers of the same model, not a contradiction; the admission readings are the

ones every feasibility statement inherits.

### *7.3. Model caveats*

*$f_{ac}$ absolute position is omitted..* The ambiguity folding is integrated about the direct-current (DC)-relative coordinate, so the absolute Doppler centre $f_{ac}$ never enters the model; only the bandwidth and the compression factor depend on squint. The measured magnitude of the omission at the design-domain edge (worst side, 45°, an extrapolated-domain reading) is 345.7 kHz (95.2 times the production PRF of 3630 Hz, and 300.6 times the 1150 Hz registration key), so the folded ambiguity structure sits far outside the band and the omission is defensible; it is disclosed regardless, and pushing the platform to its full 45° design domain must carry this number with it. A second simplification of the same family: the antenna pattern carries azimuth-frequency dependence only — there is no cross-range (elevation-plane) pattern factor — so the absolute ambiguity level is an azimuth-integration reading; no verdict in this paper rests on the unmodeled elevation-plane attenuation, and the direction contrast of Section 7.2 is carried by the bandwidth channel alone. Because the omitted term is a two-way attenuation applied to the ambiguous returns and not to the beam-centre principal image, including it could only lower the modelled ratio: the absolute ambiguity level reported here is an upper bound, and admission statements inherit that direction. Its magnitude is not quantified here.

*The C2 maneuver angle is a two-sided lower bound..* The inter-task requirement of Section 3.2 forms the maneuver angle from the difference of the tasks' start squints. This is a lower bound on the true attitude excursion from two independent sides: it omits the beam sweep during each dwell (the per-task sweep rate is 0.6686°/s at 30° incidence, 0.5599°/s at 45°, and 0.4233°/s at 60°, i.e. a 1.7–2.7° excursion across a 4 s dwell that the start-point difference never sees), and it understates the off-vertical attitude compression by 15–25%. The 25% inter-task pad is therefore a *margin argument*, not a correctness argument: it buys slack against a bound that is systematically low, and the paper never describes the

bound as an exact measure of the attitude change. One cross-reference closes this caveat: the per-task sweep rates quoted here are the antenna steering rates of Section 3.2's definition, and the antenna's own sweep during a dwell is the quantity the 1.7–2.7° excursion describes — the dwell that feeds C2 comes from the broadside-derived closure whose registered approximation status Section 3.2 records.

*$g_T$ has no external cross-validation..* The transmit-path gain term is self-derived and internally verified; the registered search for an external source found no noise-equivalent equation set in the Song platform paper (zero hits across its full text) and the textbook is not in the library with resolvable pages [22, 2]. The verification route is therefore the internal identity chain of Section 3.3 and its regression guard, and the absence of external cross-validation stands as a limitation in its own right. Consistent with the objective-level scope note of Section 3.3, no *calibrated* radiometric quantity enters the objective; the gain term shapes the resolution index and the uncalibrated relative range factor carries the range dependence, but no noise model is imported.

*The range-resolution anchor is a design constant..* $\rho_{\mathrm{rg}}$ is held at the design-anchor constant of 0.5 m, registered at 45° incidence with scene-centre semantics. Under the real-window incidence band (30–60°) a physical range resolution would vary by roughly $-18\%/+41\%$ across the band (the $\sin\eta_i$ projection of the frozen design-incidence bandwidth: $-18\%$ at 60°, $+41\%$ at 30°, each relative to the 45° anchor); the constant is a modeling choice that keeps the quality readings comparable across windows, and the magnitude narrative of any quality reading inherits this caveat.

*The quality index depends on the scheduled window..* The range factor $(R_0/R)^3$ of Section 3.3 is evaluated at the slant range of the window in which the selected start falls, and a task's windows are enumerated at different transits, hence at different slant ranges. The index of a task is therefore not a property of the task alone: moving a selection to a different transit changes its index, and with

a frozen mode assignment that change is one of the quantities a schedule can still choose. The variation is not corrected anywhere in this paper, and it is the same non-normalisation that leaves index levels incomparable across families (Section 5.2).

*Two quality calibers, and the one this paper reports..* The fulfillment ratio is computed on the library caliber, while the delivered variant $f_{Q,\mathrm{deliv}}$ is computed on a delivered caliber; the two are different quantities at different calibers, are never reported side by side in one table, and are never combined into a ratio of one to the other. The reading this paper reports is the library-caliber one, and its limitation is that it measures a task's realised value against its own zero-squint optimum over its own windows — a within-task, within-family quantity. It is not a calibrated radiometric statement, it does not compare across families (the units expand scene-dependently, Section 5.2), and it is not the delivered resolution of the schedule. The delivered variant carries a limitation in the other direction: it has no stripmap domain at all, because stripmap delivers the library floor and a task-level mode-parameter request on that mode raises, so the variant is undefined for every task the control family serves.

*Per-mode calibers are not economically comparable..* The spotlight admission in the mode library is defined per point target while the sliding-spotlight scenes are scene-level, and the scheduler consumes the scene-level dwell; the two calibers are registered as asymmetric. Any statement that trades dwell economy across modes — a spotlight dwell against a sliding-spotlight dwell as if they purchased the same thing — is outside what the model supports, and none is made.

### *7.4. Design semantics and pre-answers*

*Why not jointly optimize mode and time?.* Because the mode is a product-order property, not a scheduler preference. The mode follows from the ordered (resolution requirement, scene size) pair by the coarsest-sufficient library table; making it a decision gene would let the optimizer silently upgrade or downgrade the product the customer ordered. The sliding factor is likewise a system

parameter. The evidence this paper can offer for the joint formulation being unnecessary here is narrower than that argument needs, and stating it precisely matters: the coverage response is a construction response (Section 4.5), so in this scenario world a unit's outcome is set before the search runs, and a mode gene would add decision variables to a search whose measured contribution is two units out of 260; whether a mode gene would change the served set under a different demand mix is untested. Joint mode-and-time allocation remains future work for regimes where that activation exists.

*The reading is single-budget..* The curve of Section 6 is read at one registered budget, and a future budget extension can move it: nothing in the registered design forbids a re-run at a larger budget, and if a larger budget changed the served sets, the registered criteria of Section 5.3 would be re-measured rather than assumed. In practice the budget was fixed by a registered decision — raising it was judged to buy a different search rather than a different construction, since the construction face carries the response (Section 4.5) — so the extension is available on paper and priced out by policy. That asymmetry is disclosed rather than hidden, and it bounds every reading in this paper.

## 8. Conclusion

This paper put the order resolution into the scheduling constraint set and measured what tightening it costs. On the platform leg, the imaging mode is resolved from the product order and its parameters follow the order per task, so the delivered resolution equals the order at the selected start by construction; the dwell is then an endogenous consequence of the mode, the order and the selected start, and the maneuver-coupled feasibility chain prices every schedule in physically meaningful time. The quality index states a platform capability and supplies a reported reading — a fulfillment ratio on the library caliber — and it carries no selection pressure: coverage is the only objective, and a unit's front is a tie-set on it.

The result is a coverage–demand curve over a frozen nine-point grid, read under a fixed mode assignment. The 100-drawn / 99-admitted family is where the tightening bites, from 0.4827 at the tightest feasible point to 0.9900 at the coarsest, and it pays 4.15 percentage points at the order point against its own saturation. The 50-drawn / 46-admitted family is nearly flat and pays 2.36 points. The single-orbit control is a floor step: it serves the whole priority mass from the order point upwards, reads 0.5398 at the next point down, and has no feasible individual at all at the tightest point — an infeasibility record, not a coverage of zero. Two of the three frozen shape criteria fail on the 50-task family, and both failures are reported as this world's answer; the tight-arm penalty of the previous world is withdrawn rather than quietly carried forward. Over the grid the tie set serves 477,308 task instances with no violation recorded relative to the evaluator, and the anchor already attains the unit maximum in 258 of 260 units — which is what makes the response a construction result rather than a search finding.

The claims stay inside those bounds. They are order sensitivity under a frozen mode assignment, at one registered budget, with the capacity law stated in caliber and without a number. The sensor-layer question of resolution against swath, the joint mode-and-time formulation, the recomputation of the serial bound under this scenario world, and an operational-cost evaluation are future work rather than results.

## CRediT authorship contribution statement

**Liu Shuhao:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization.

## Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported

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**Data availability**

The promoted experiment artifacts (scenario manifests, per-unit pilot records across the three budget tiers, and the convergence-calibration registration), the preregistration file with its dated amendment chain, the caliber registry, and the verification script that recomputes the registered verdicts from those artifacts are available from the corresponding author. No proprietary data were used; the scenario world is generated from a registered synthetic template.

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